Mastering Power BI

Power BI Mini Project

A Step-by-Step Walkthrough from Data Preparation to Data Modeling, Visualization and Publishing to Show You What Power BI Is All About

Get Data (Multiple Sources) › Transform & Clean › Model Your Data › Create Visualizations › Publish & Share



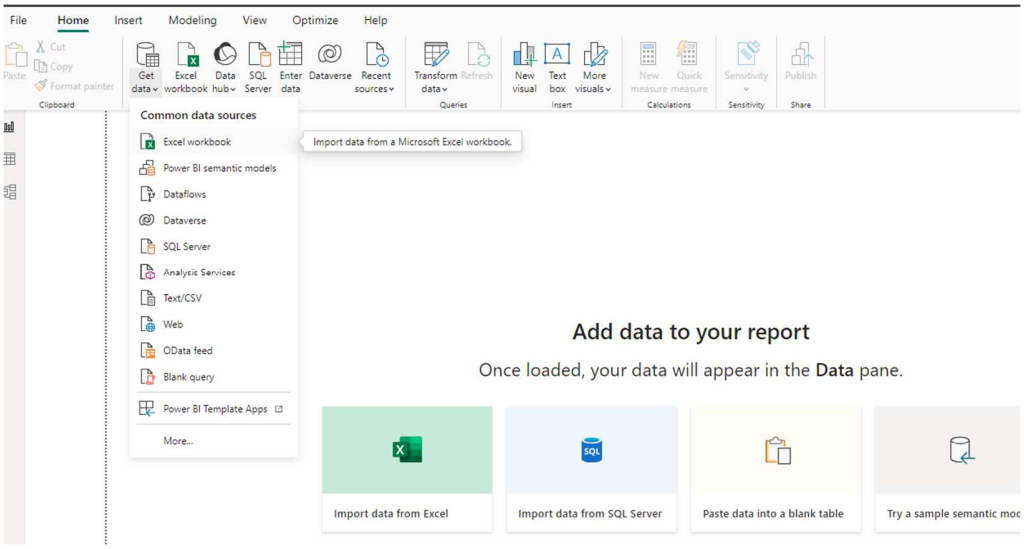
Learn by Doing | Build a Real Project | Understand the Full Process | Gain Confidence with Power BI

Said Fawzy

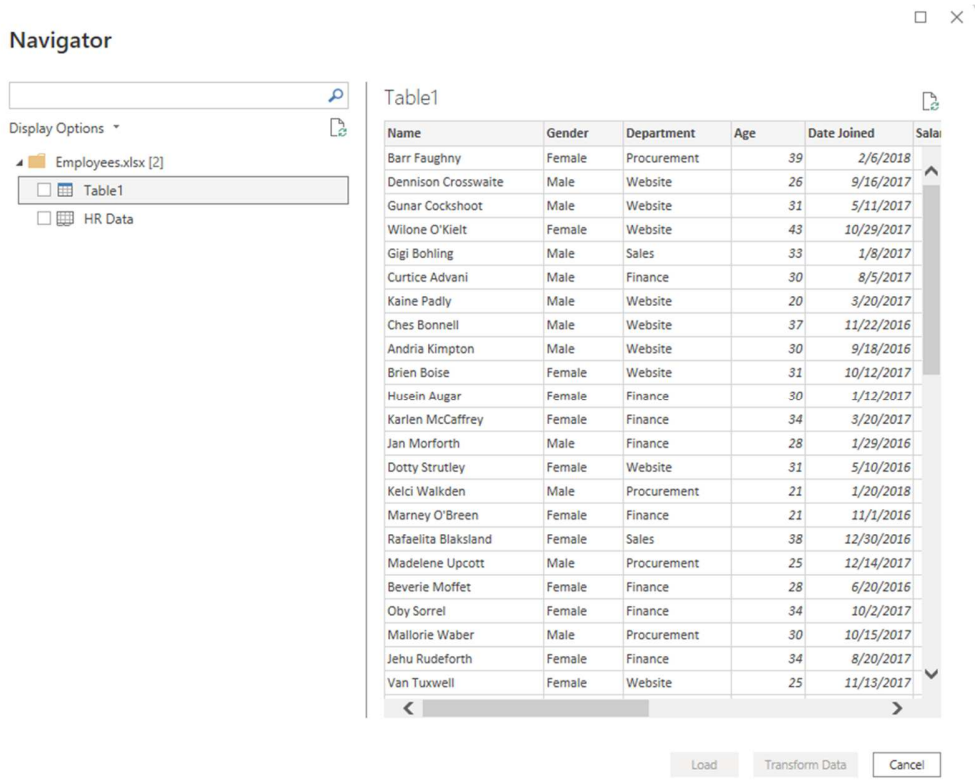
Power BI Mini Project

A: Getting Data

1. Use the Excel file: **Employees.xlsx** in Lab folder.
2. Explore the data in the workbook and .
3. Notice you have a worksheet with the Name **HR** and a table with name **table1**.
4. Explore columns you have.
5. Open Power BI.
6. Close the **splash** screen.
7. From **Home** Ribbon in **Data** Group Select **Get Data**.

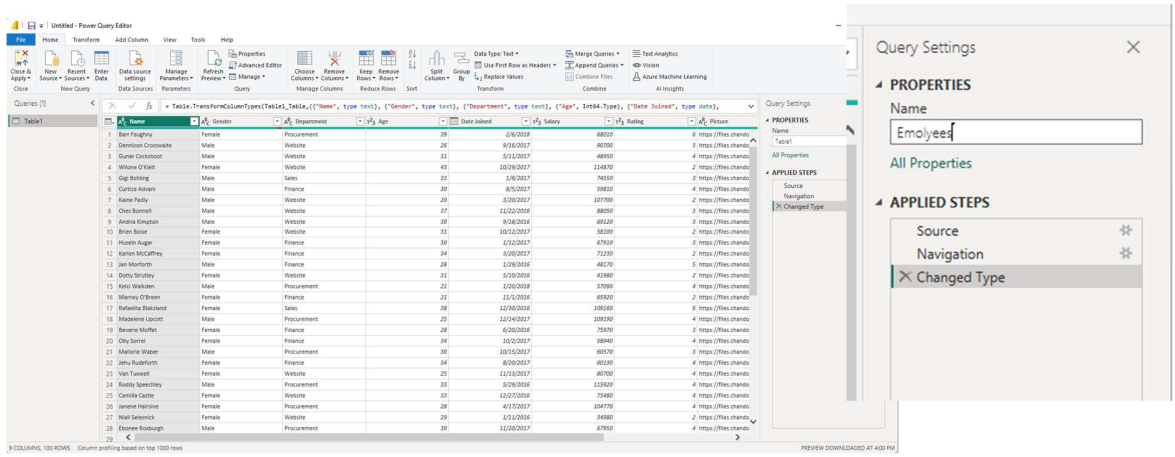


8. Select Excel Workbook.
9. Browse to your Employees workbook to open.
10. In the **Navigation Pane** notice that on the left you have the tables that are available in the source and when you click a table you get a preview on the right side.
11. You have a sheet and a table with the same date.
12. Select **table1**.



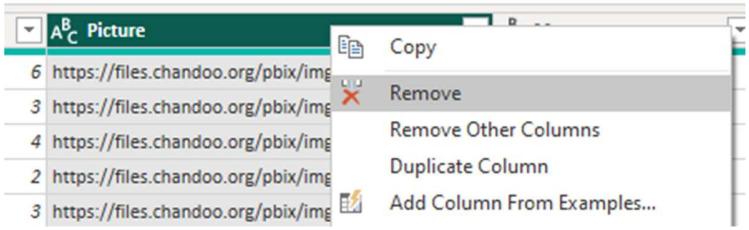
13. You can select Loade to **load** to your data model directly.
14. But select **Transform** to go to power query



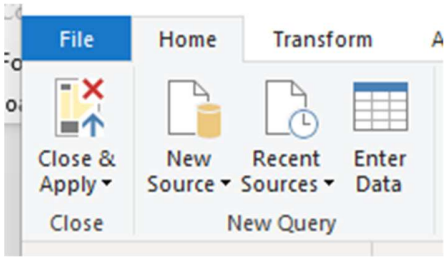


15. The Power Query is opened in a separate window.
16. Explore Power Query.
17. You have your table in the middle of the screen.
18. You have Ribbons to help you to transform data.
19. On the left you have Queries Pane.
20. In right you have Query settings with Properties and the Applied Steps.
21. Change the name of your Query to **Employees**.

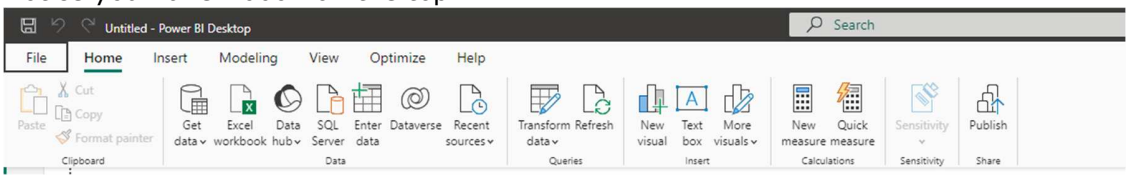
22. We do not need the Picture column.
23. Right click then choose to remove.



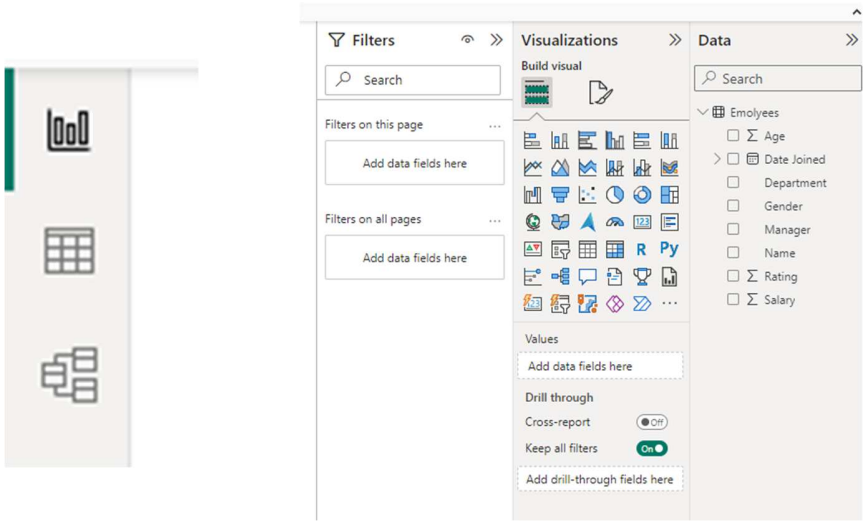
24. Notice that you have the step added to the **Applied steps**.
25. You can undo what you have done by just deleting the step from the Applied steps.
26. Do and redo your removing column.
27. You can now click close and Apply in the Home tab to close power query and get back to the Power BI.



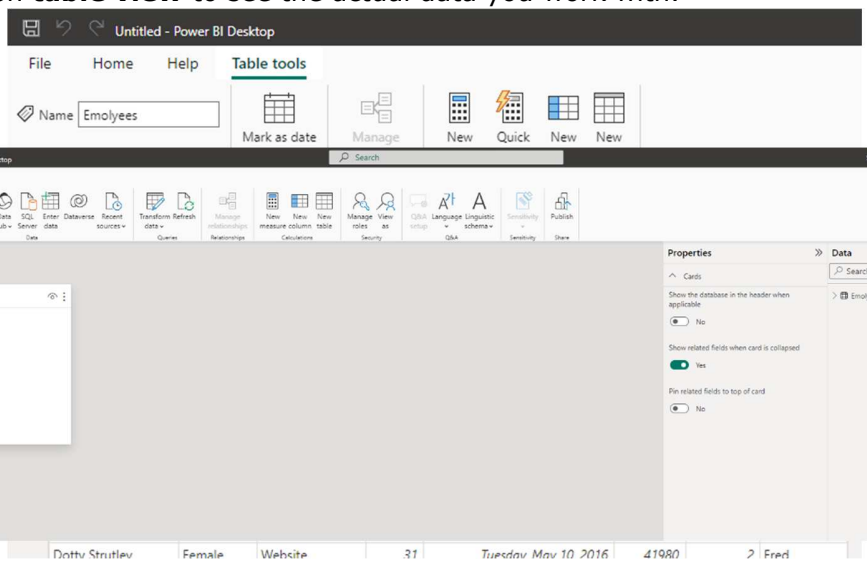
28. You get back to your Power BI File.
29. Notice you have ribbon on the top .



30. You have Pans on the right : Data, Visualization , and Filter.
31. And on the left you have the three icons of the 3 view of power BI which are : **Report View ,table View and Model view.**



32. Report view is where you do your visulization.
33. Click on **table view** to see the actual data you work with.
34. Click on **Model view** to see Your data model



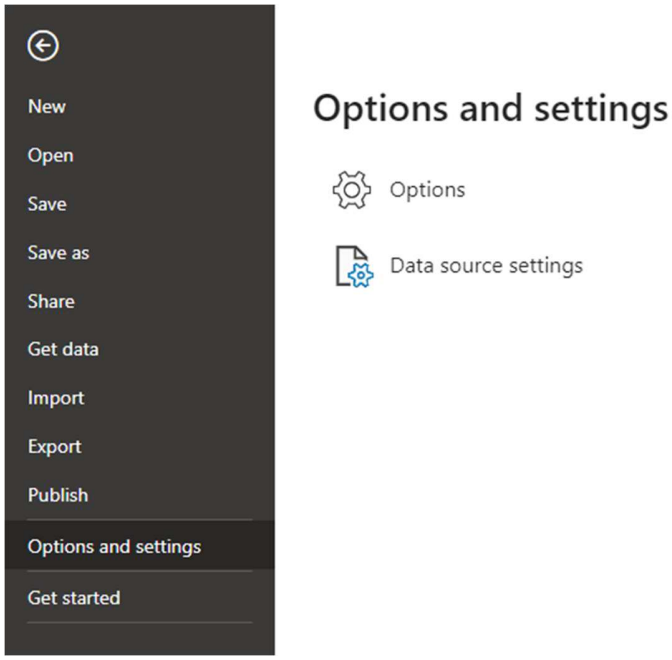
35. Notice is only made of one table this time.

36. Go back to Report View.
37. Notice that is an old View of Power BI.

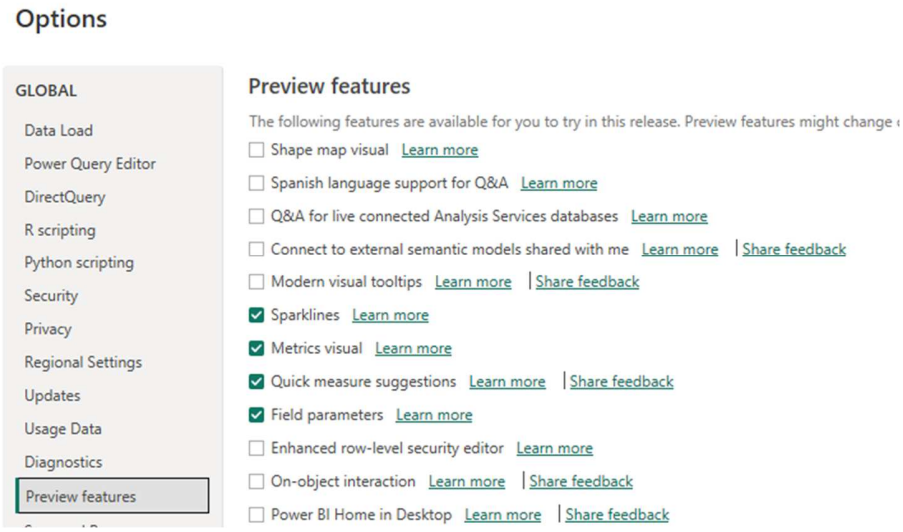
B: Change Power BI Settings to on Object Interaction

38. Let us move our view to On Object Interaction View.
39. First Save your File as **My First Power BI Report.pbix.**

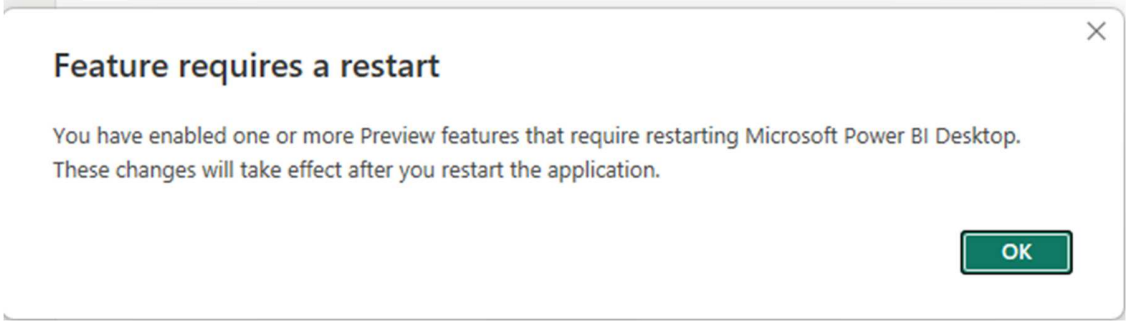
40. Go to File ➔ Option and settings ➔ Options.



41. In Google ➔ Preview Features ➔ select **On-Object Interaction**

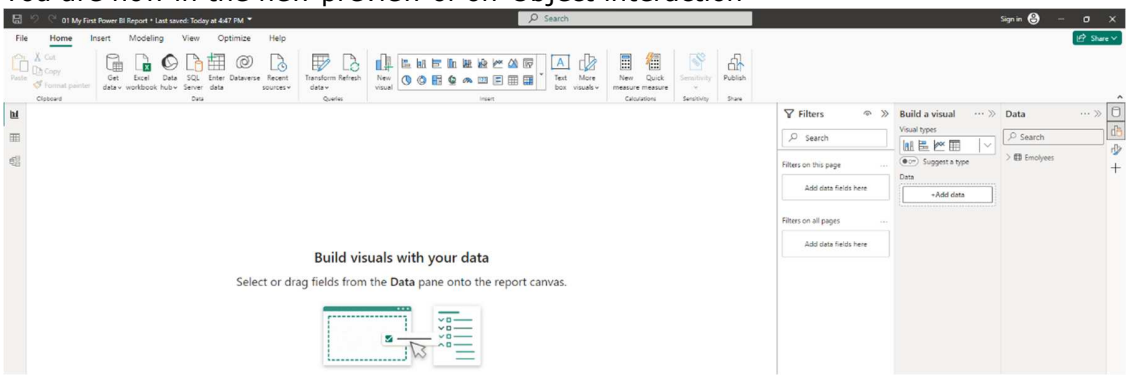


42. You got a restart message requirement.



43. Close and reopen Power BI.

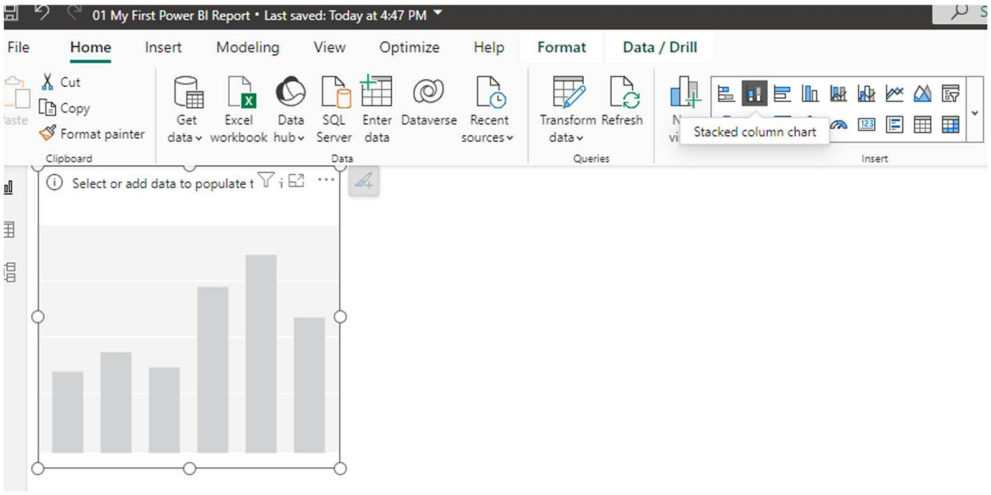
44. You are now in the new preview of on-Object interaction



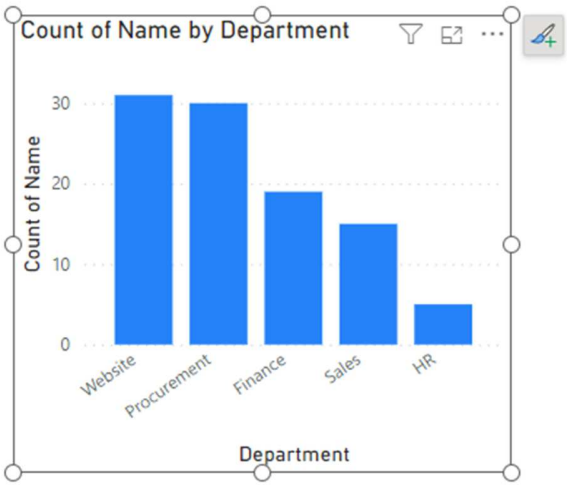
45. Notice that Visuals are now on the top and you have a new interface.

C: Creating Bar Chart

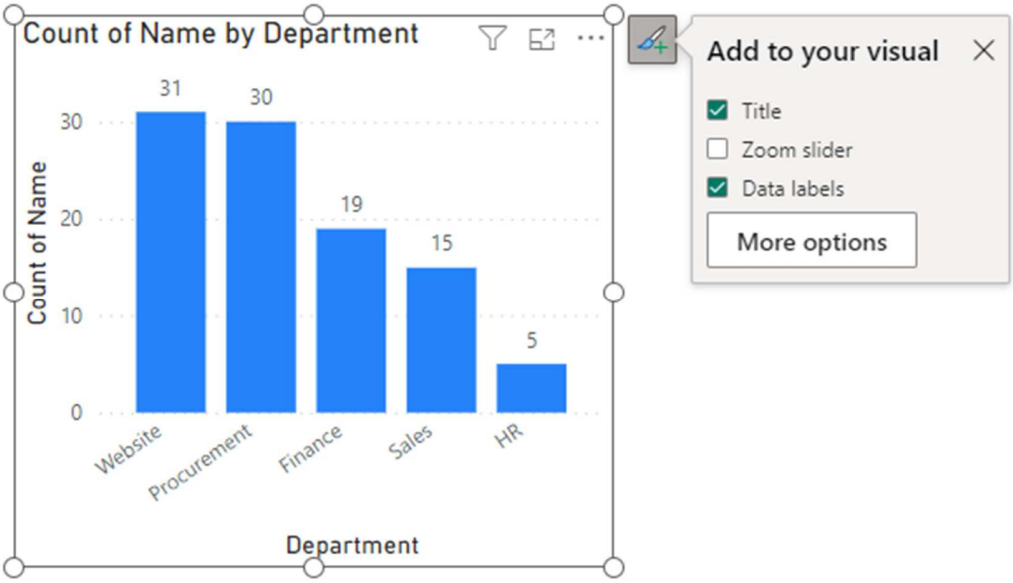
46. Select Stacked column visual to appear in your report Canvas.



47. Your boss wants to know How many employees in each department.
48. Drag Department from data pane to X-axis in Build pane , and drag name to Y-axis (it will then count).
49. You have now the count of employees in each department.



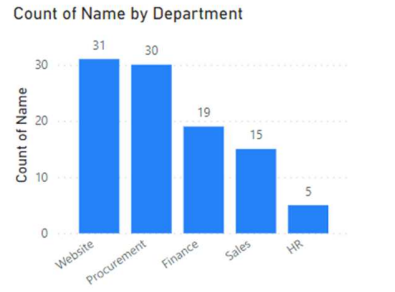
50. Click on the icon on the right of the Visual (add and remove to your visual) and select Data Labels.
51. The Number of employees is now Appear on top of each column.

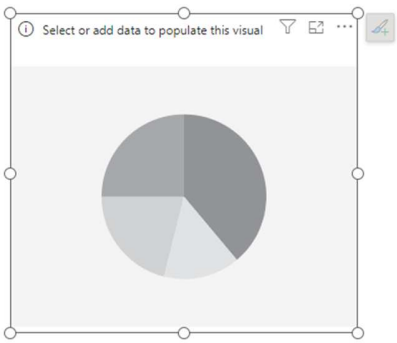


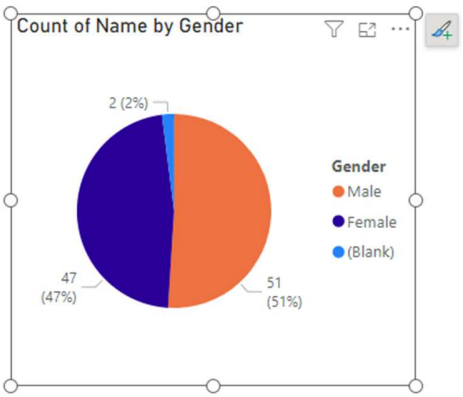
D: Create Pie Chart

52. We want to what is the ratio between men and women in the company
53. The suitable visual here is Pie Chart.
54. First Click on a free space in Canvas.
55. Then click on Pie from the visual so it is in your Report.

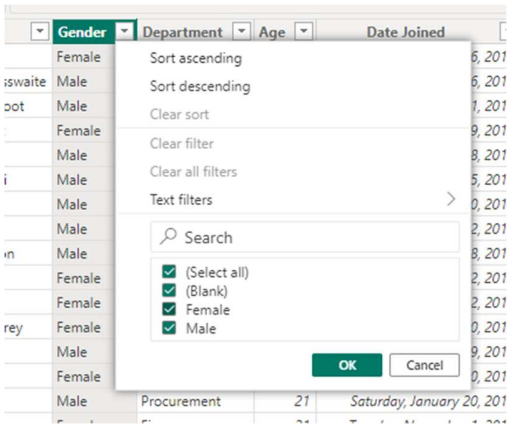
56. Drag Ginder to legend and Names to Values, so Power BI count them.

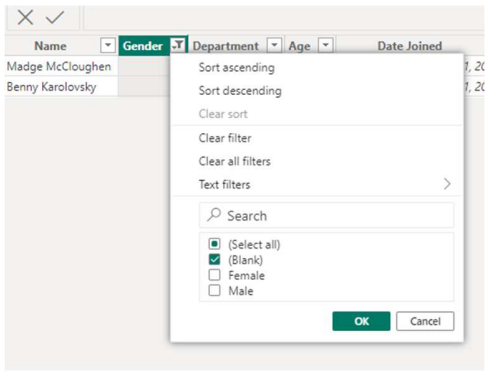






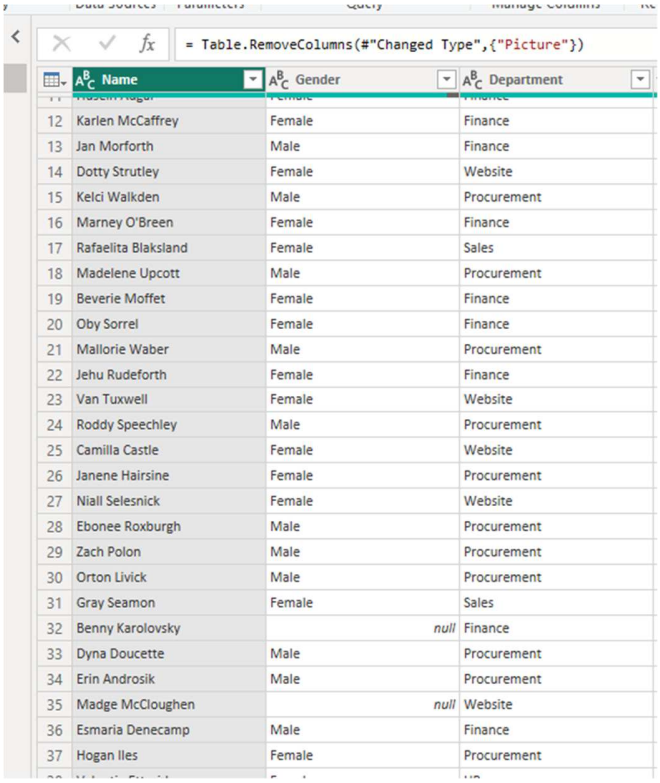
57. Notice you have 3 Gender in the legend **Male**, **female** and **Blank**.
58. That means you have a missing value in this column.
59. Go to **table view** and check the **Gender** column.



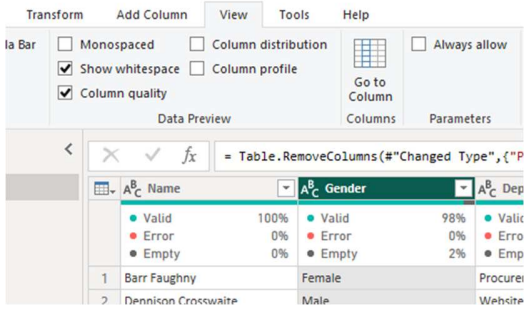


60. Notice you have blank value.
61. You can filter the view to see them.
62. Try to change the value of those two rows, you cannot.
63. To transform the data, you must use power query editor.
64. From the **home** tab in data ribbon chose **Transform data** to open power query again.

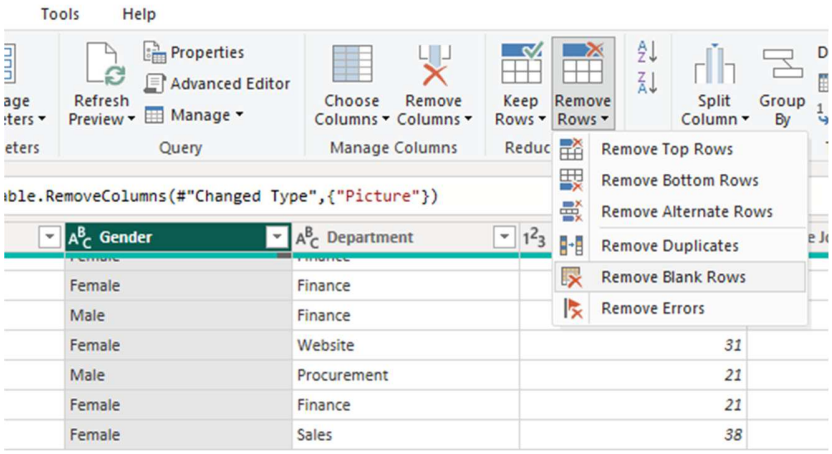
65. Notice you have two null values in the column.
66. Also, if you go to **view** tab ➔ **Data Preview** ➔ check **column quality**.
67. You get the % of **valid, error** and **empty** values.



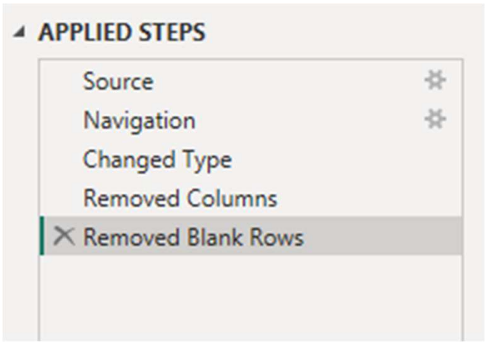
	Name	Gender	Department
12	Karlen McCaffrey	Female	Finance
13	Jan Morforth	Male	Finance
14	Dotty Strutley	Female	Website
15	Kelci Walkden	Male	Procurement
16	Marney O'Breen	Female	Finance
17	Rafaelita Blaksland	Female	Sales
18	Madelene Upcott	Male	Procurement
19	Beverie Moffet	Female	Finance
20	Oby Sorrel	Female	Finance
21	Mallorie Waber	Male	Procurement
22	Jehu Rudeforth	Female	Finance
23	Van Tuxwell	Female	Website
24	Roddy Speechley	Male	Procurement
25	Camilla Castle	Female	Website
26	Janene Hairsine	Female	Procurement
27	Niall Selesnick	Female	Website
28	Ebonee Roxburgh	Male	Procurement
29	Zach Polon	Male	Procurement
30	Orton Livick	Male	Procurement
31	Gray Seamon	Female	Sales
32	Benny Karolovsky	*null*	Finance
33	Dyna Doucette	Male	Procurement
34	Erin Androsik	Male	Procurement
35	Madge McCloughen	*null*	Website
36	Esmaria Denecamp	Male	Finance
37	Hogan Iles	Female	Procurement



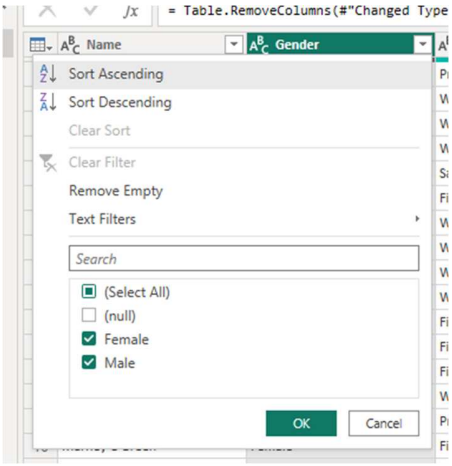
68. You must decide now.
69. You can delete those rows.
70. Click Home ➔ Remove Rows ➔Remove Blank Rows.

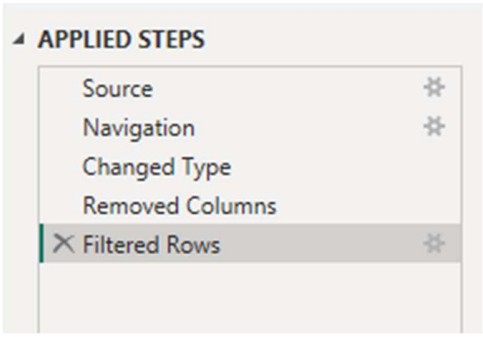


71. It is not a good decision.
72. Go and delete that step from applied steps pan in the right.



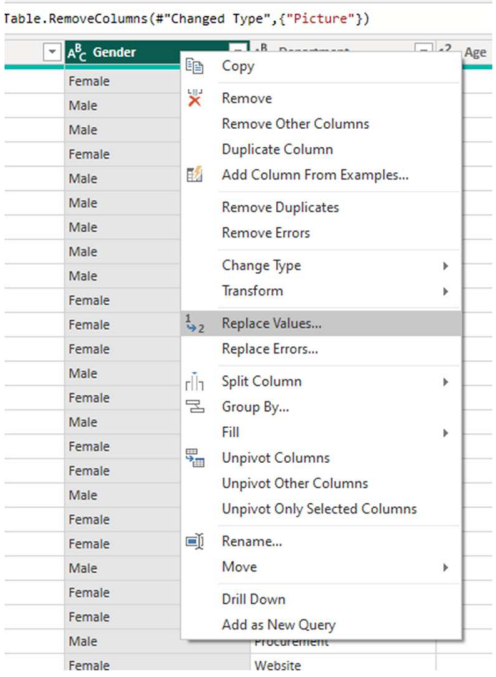
73. You can do the same thing by filtering the value in the Gender column.



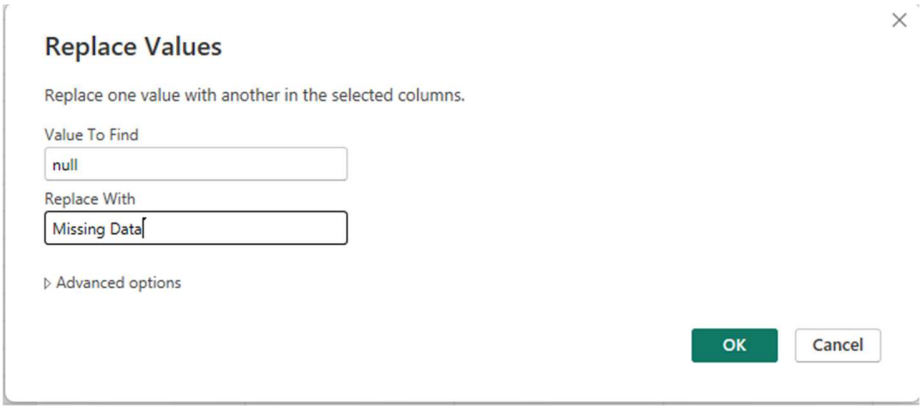


74. Also, it is not a good decision, go and delete this step too.
75. The best thing is to put value in this blank like "**missing data"**.

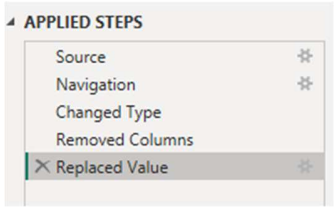
76. Right click the Gender column and choose Replace Value.

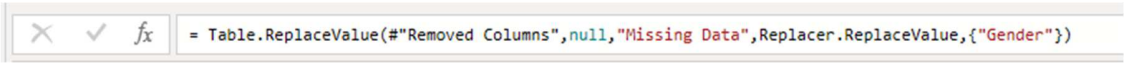


77. In the replace value dialogue box replace null with missing data.



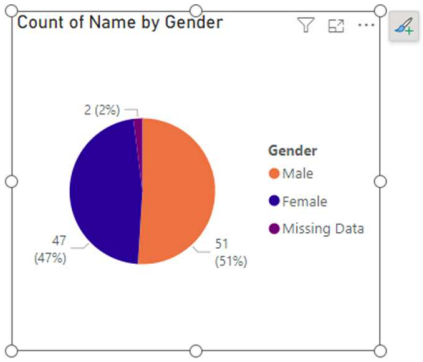
78. It is a chance to see the applied step pan.
79. Click every step and watch the corresponding **M language** line in the formula bar.
80. You can also go to **Home** tab ➔ **Query** Group ➔ **advanced editor**.





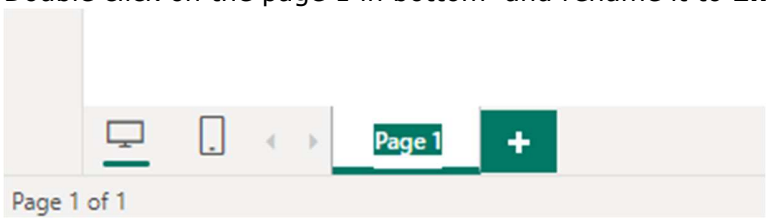
81. You can here see and edit the steps manually in **M language**.
82. From **home** tab click **Close & Apply** to go back to power BI.
83. Notice now it appears in **table view** and **repot view** in the Pie chart.

Niall Selesnick	Female	Website
Ebonee Roxburgh	Male	Procurement
Zach Polon	Male	Procurement
Orton Livick	Male	Procurement
Gray Seamon	Female	Sales
Benny Karolovsky	Missing Data	Finance
Dyna Doucette	Male	Procurement
Erin Androsik	Male	Procurement
Madge McCloughen	Missing Data	Website
Esmaria Denecamp	Male	Finance

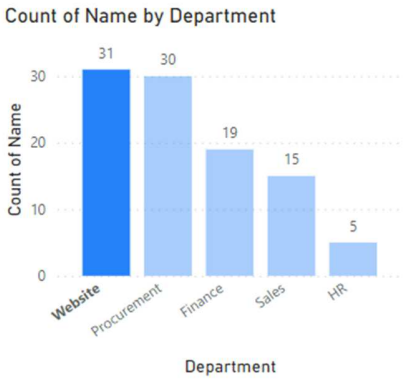


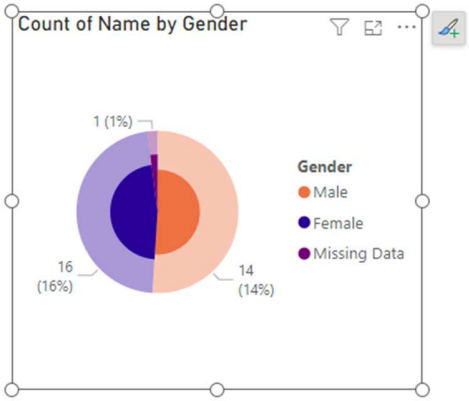
E: Visuals interactivity

84. Now you have two visuals in your report.
85. It is Page one of your report.
86. Double click on the page 1 in bottom and rename it to **Employees**.



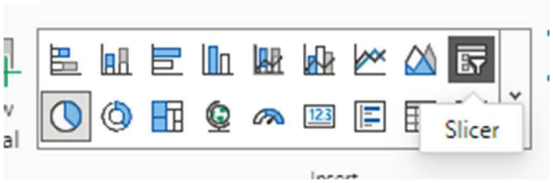
87. You can also click on + button and add a new page.
88. Click on Column Website department in the column chart.
89. You will see that pie chart reflects that and show only value of that department only.
90. Go and click on every department to see their values in pie chart.

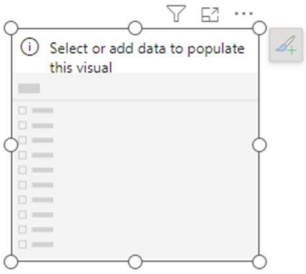




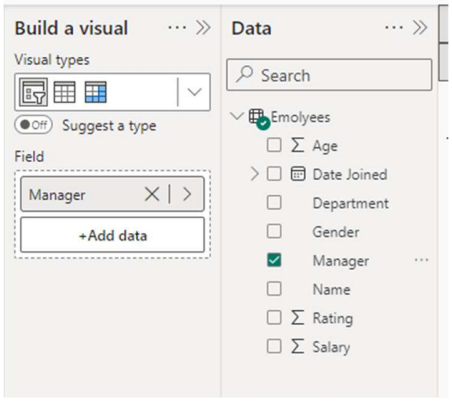
91. Now we want to filter by manager

92. You can add visual slicer and add manger to it.

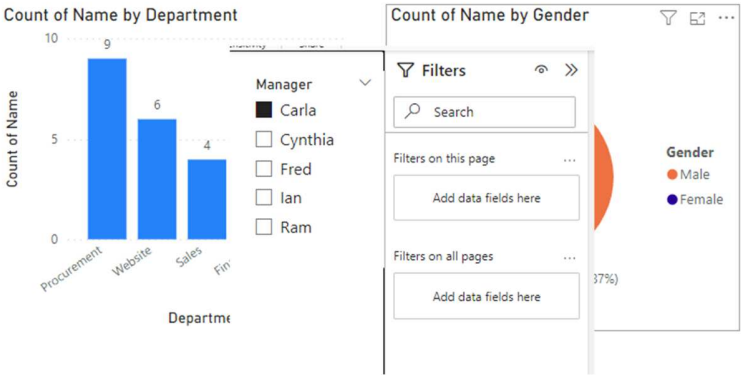


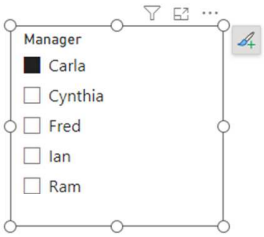


93. Make sure you always click on empty part of canvas before you add any new visual.
94. Now you can filter the two visuals with your new slicer.









95. Notice you can do that by using filter pane to save space, but slicers are so intuitive for the users of your report.

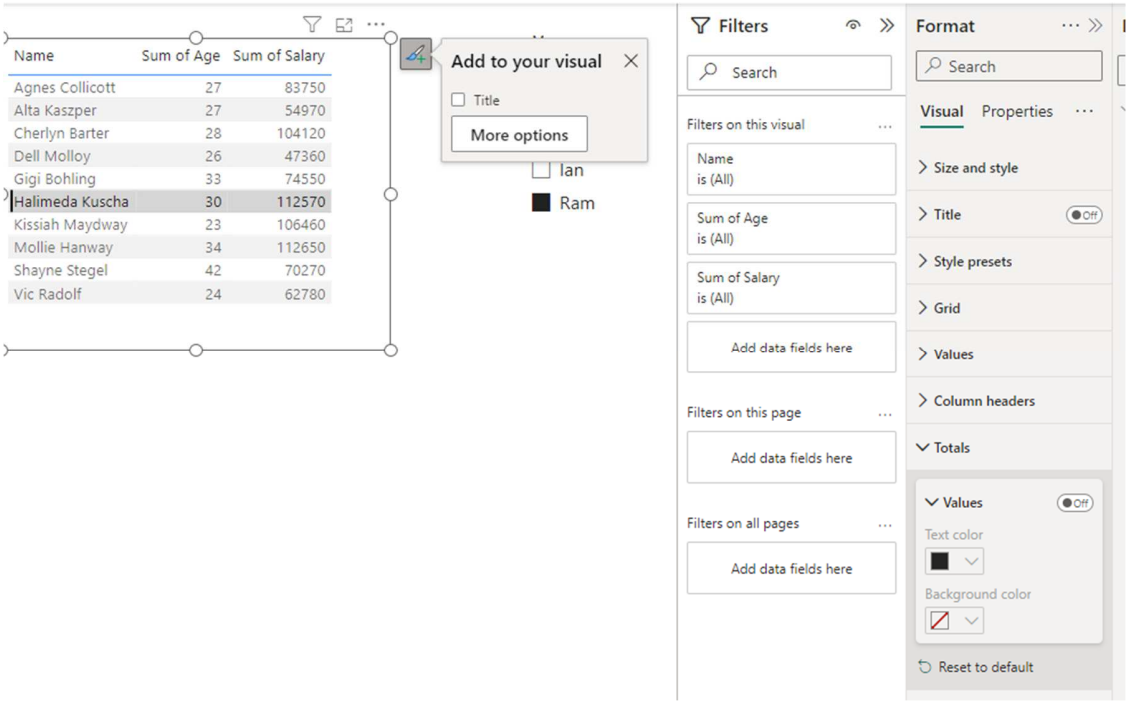
F: Creating table and format results

1. Your boss liked your report, but it asked you to see the employees of each manager.
2. Add a table to your report
3. In Data pane click on Name, age, rating and salary.

4. You must now have your table like this.

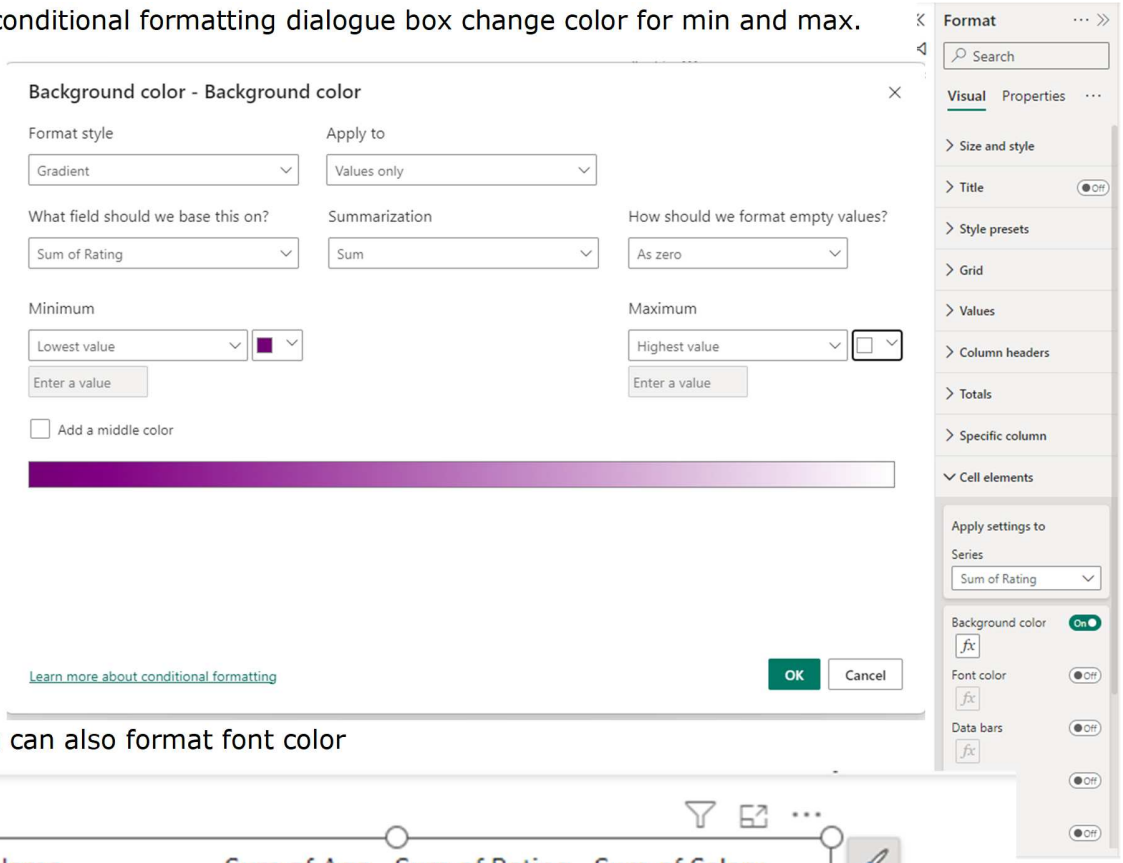
Name	Sum of Age	Sum of Salary
Agnes Collicott	27	83750
Alta Kaszper	27	54970
Cherlyn Barter	28	104120
Dell Molloy	26	47360
Gigi Bohling	33	74550
Halimeda Kuscha	30	112570
Kissiah Maydway	23	106460
Mollie Hanway	34	112650
Shayne Stegel	42	70270
Vic Radolf	24	62780
Total	**294**	**829480**

5. Go to icon in right of table (format) ➔More Options ➔Visual tab ➔Totals ➔ Values and make it **off**.

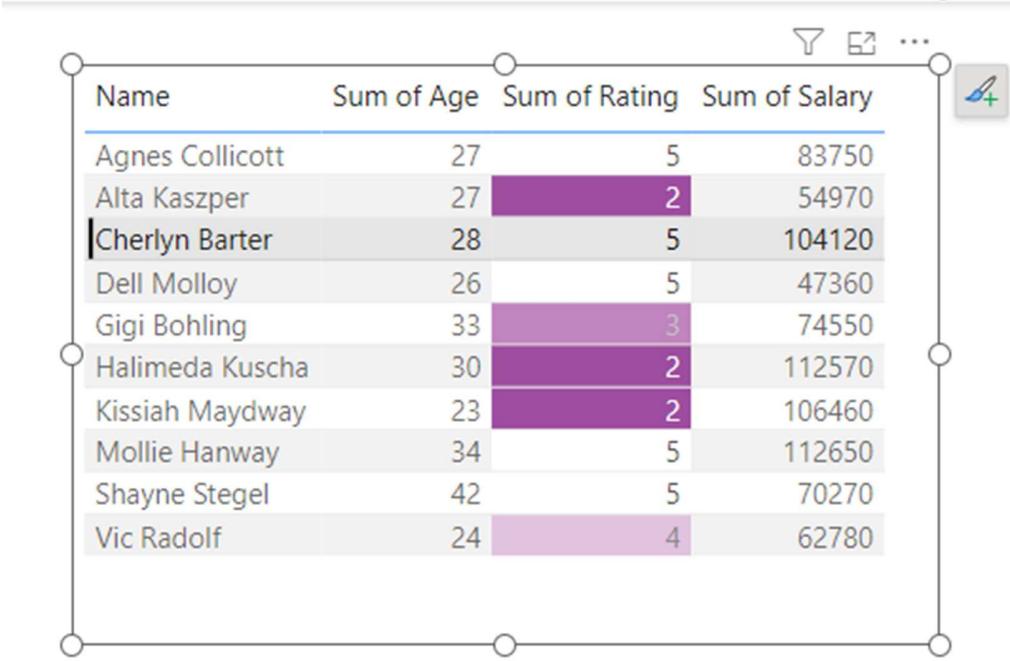


6. Notice you have search bar in format pane you can search for what you want.
7. Your boss is happy now he can see every manager employee.
8. He asked you to see which employees have low ratings.
9. He wanted to having conditional formatting for rating column.
10. Go to Format pan ➔ Visual➔Cell elements➔Apply setting to ➔ choose rating ➔Make back ground color **on** and thin click the fx icon.

11. In conditional formatting dialogue box change color for min and max.



12. You can also format font color

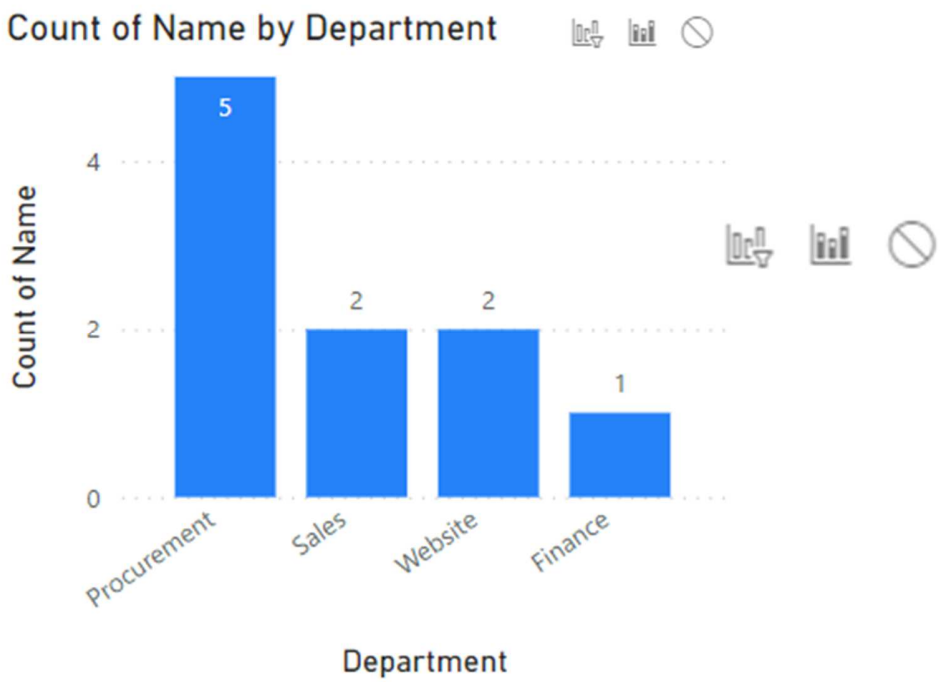


Name	Sum of Age	Sum of Rating	Sum of Salary
Agnes Collicott	27	5	83750
Alta Kaszper	27	2	54970
Cherlyn Barter	28	5	104120
Dell Molloy	26	5	47360
Gigi Bohling	33	3	74550
Halimeda Kuscha	30	2	112570
Kissiah Maydway	23	2	106460
Mollie Hanway	34	5	112650
Shayne Stegel	42	5	70270
Vic Radolf	24	4	62780

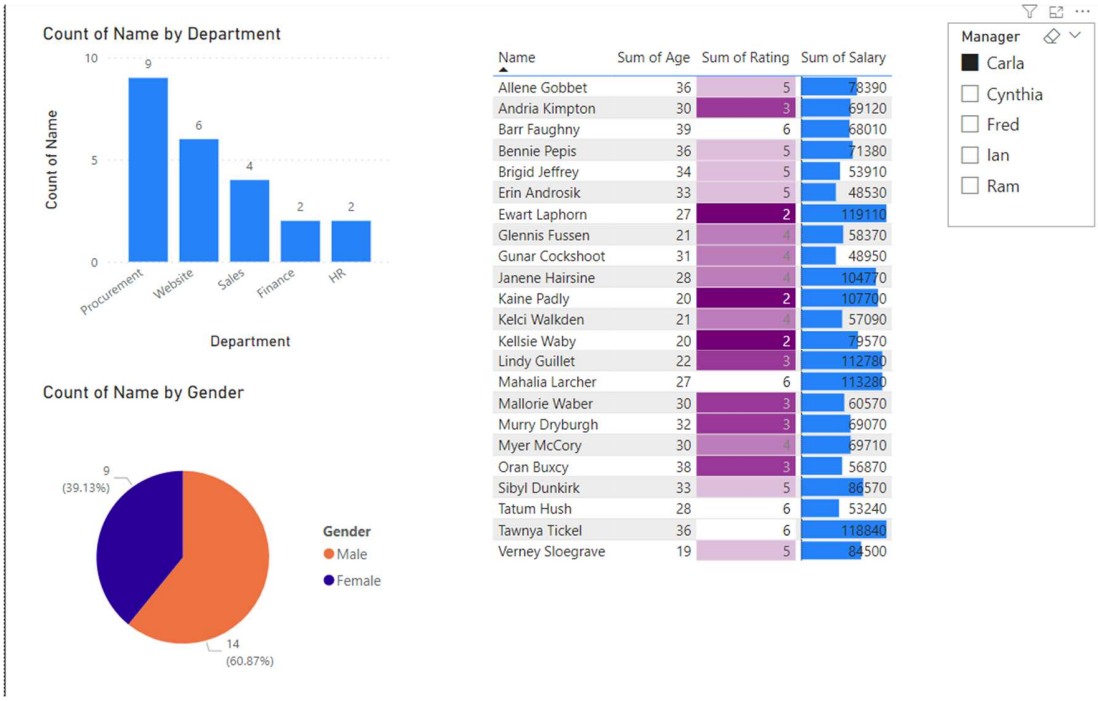
13. Your boss wanted to highlight the high salary.
14. Chose salary this time and use the data bars options and make it on.
15. You must have your table now like this

Name	Sum of Age	Sum of Rating	Sum of Salary
Agnes Collicott	27	5	83750
Alta Kaszper	27	2	54970
Cherlyn Barter	28	5	104120
Dell Molloy	26	5	47360
Gigi Bohling	33	3	74550
Halimeda Kuscha	30	2	112570
Kissiah Maydway	23	2	106460
Mollie Hanway	34	5	112650
Shayne Stegel	42	5	70270
Vic Radolf	24	4	62780

16. Notice that table also when you click on an employee it filters the other two charts which has no meaning.
17. Stop that as follow as you select the table chart:
    - Go to Format tab ➔Indicators group ➔ edit indicators.



    - This shows new icons on top right of other visuals.
    - They are Filter, Highlight, **non**.
    - Chose non for all other visuals.
    - Click edit indicators again to remove these icons.
18. Now your final report like this:

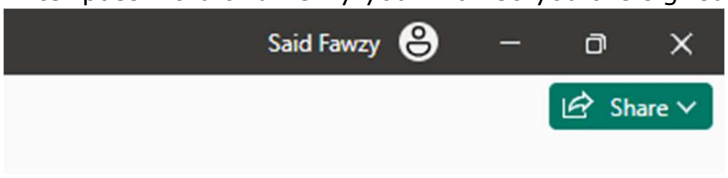


Name	Sum of Age	Sum of Rating	Sum of Salary
Allene Gobbet	36	5	78390
Andria Kimpton	30	3	69120
Barr Faughny	39	6	68010
Bennie Pepis	36	5	71380
Brigid Jeffrey	34	5	53910
Erin Androsik	33	5	48530
Ewart Laphorn	27	2	119110
Glennis Fussen	21	4	58370
Gunar Cockshoot	31	4	48950
Janene Hairsine	28	4	104770
Kaine Padly	20	2	107700
Kelci Walkden	21	4	57090
Kellsie Waby	20	2	79570
Lindy Guillet	22	3	112780
Mahalia Larcher	27	6	113280
Mallorie Waber	30	3	60570
Murry Dryburgh	32	3	69070
Myer McCory	30	4	69710
Oran Buxcy	38	3	56870
Sibyl Dunkirk	33	5	86570
Tatum Hush	28	6	53240
Tawnya Tickel	36	6	118840
Verney Sloegrave	19	5	84500

19. You can click any column Header to sort.
20. Save your work.

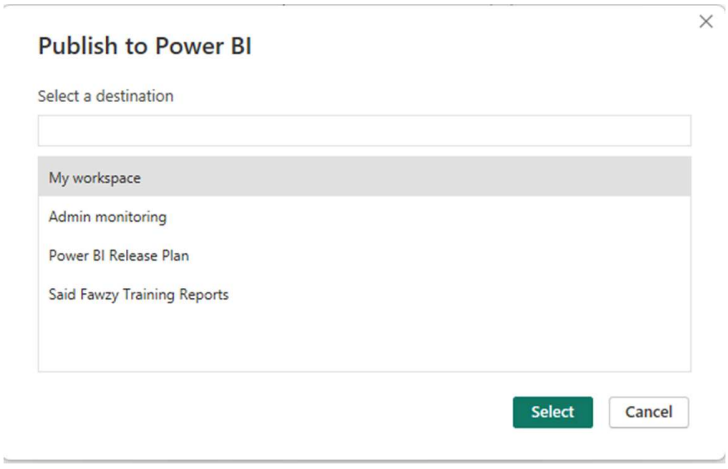
G: Publish and share your report.

21. Click sign in in top right of the screen.
22. Enter your business mail or developer mail and press continue.
23. Enter pass word and verify your mail so you are signed in

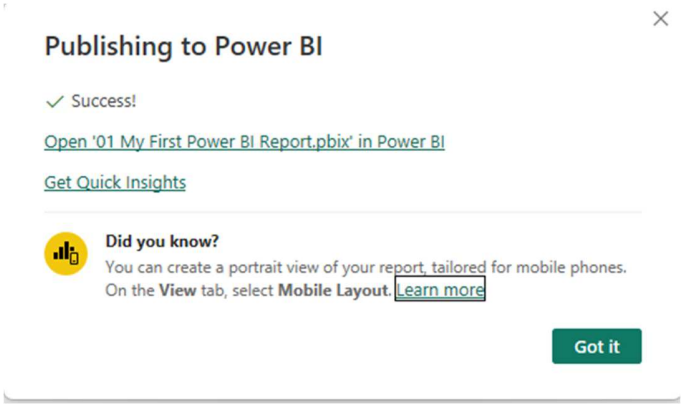


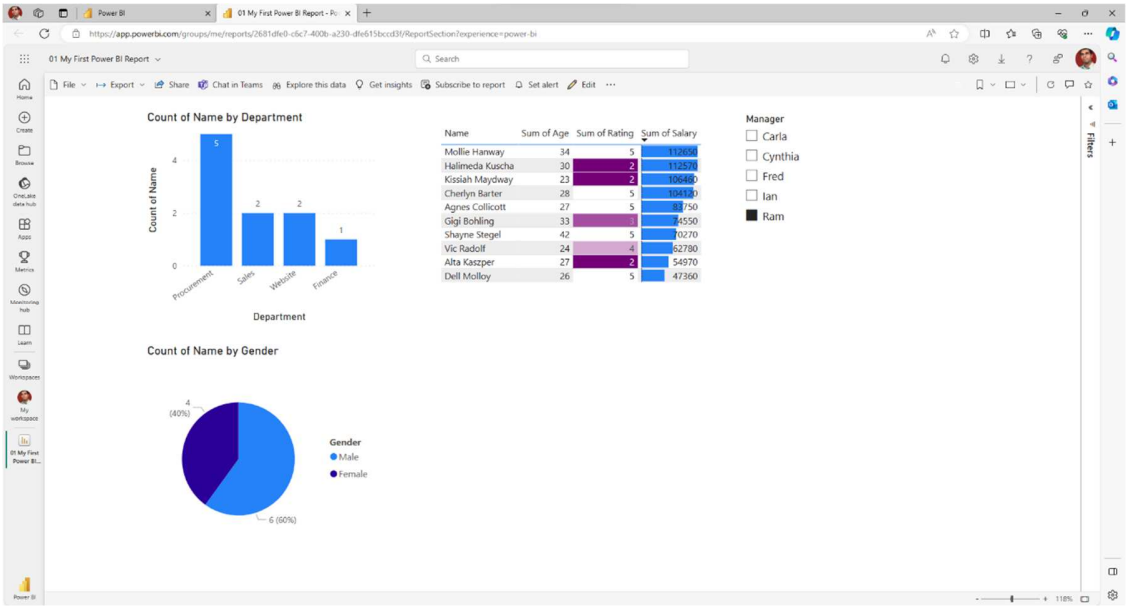
24. on home tab click **publish**.
25. Select your destination workspace





26. Then click the link to go to your report





27. Now your report is published in your workspace, and you can share it if you have pro account